

ORIGINAL ARTICLE

Evaluation of Transcutaneous Electroacupoint Stimulation with the Train-of-four Mode for Preventing Nausea and Vomiting after Laparoscopic Cholecystectomy

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ABSTRACT **Objective:** To evaluate the efficacy of transcutaneous electroacupoint stimulation with a train-of-four (TOF) mode for the prevention of postoperative nausea and vomiting (PONV) in the patients undergoing laparoscopic cholecystectomy. **Methods:** Ninety-six ASA Grade I–II patients scheduled for laparoscopic cholecystectomy were randomized into Neiguan (P6) electroacupoint stimulation group (treated group) and a placebo control group (placement of electrodes without electroacupoint stimulation). The anesthetic regimen was standardized by needling at Neiguan on the left side and connecting the TOF peripheral nerve stimulator. The incidence of nausea, vomiting, severity, antiemetic dosage and the degree of pain were assessed at 0, 60, 120 min, and 24 h after surgery. **Results:** The incidence of nausea and vomiting, the dose of antiemetics and the occurrence of severe nausea were all significantly lower in the treated group compared with the control group and the score for pain was obviously reduced in patients of the treated group at 24 h post-operation ($P<0.05$ or $P<0.01$). **Conclusion:** Transcutaneous electroacupoint stimulation at P6 with the TOF mode could reduce the incidence and severity of nausea and vomiting with analgesic effects.

KEY WORDS transcutaneous acupoint electrical stimulation, train-of-four stimulation, postoperative nausea and vomiting, laparoscopic cholecystectomy

Postoperative nausea and vomiting (PONV) is a common complication in patients undergoing laparoscopic surgery⁽¹⁻³⁾. Patients may feel extreme discomfort and may have serious complications such as dehydration, electrolyte disturbances, venous hypertension and subcutaneous hemorrhage which can affect the quality of the surgical outcome^(1,2). The incidence of PONV, which can be as high as 53%-80%^(1,2,4), is influenced by various patient-related factors including the type of surgery, anesthesia techniques and drugs used as well as postoperative factors such as pain, dizziness, changes of posture and usage of opioids, etc⁽²⁾. Various drugs have been found to be effective in decreasing PONV but have their own adverse effects. Selective serotonin antagonists are now the main therapy in the prevention and treatment of PONV, but there are some limitations. The antiemetic effect of Neiguan (P6) stimulation by various methods has been reported, such as acupuncture, electroacupuncture, transcutaneous electrical nerve stimulation, acupoint stimulation, and acupressure⁽⁵⁾. Recently, Khan, et al. described

the use of the conventional peripheral nerve stimulator (PNS) to stimulate P6 with the train-of-four (TOF) mode for the prevention of PONV after surgery⁽⁶⁾. This seems to be a simple, inexpensive and readily available method, but its efficacy has not been fully evaluated. The objective of the current study is to evaluate the efficacy of PNS at P6 to prevent PONV in patients undergoing laparoscopic cholecystectomy under general anesthesia.

METHODS

Subjects

From June 2006 to July 2007, a total of 96 patients with chronic cholecystitis scheduled for laparoscopic cholecystectomy were enrolled in this randomized, prospective, double-blinded study. Patients who were pregnant, experiencing menstrual

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Table 1. Demographic and Preoperative Physiologic Values

Group	Case	Gender (Male/Female)	Age (Year, $\bar{x} \pm s$)	Boby weight (kg, $\bar{x} \pm s$)	ASA grade (I / II , Case)	Previous PONV (Case)
Treated	48	26/22	42 $\pm$ 18	61 $\pm$ 12	40/8	8
Control	48	25/23	40 $\pm$ 19	58 $\pm$ 13	38/10	7

symptoms, using a permanent cardiac pace-maker, have had acupuncture therapies before surgery, had received antiemetics or experienced nausea, vomiting, or retching within 24 h of surgery were all excluded.

The patients, aged 18 to 60 years with ASA Grade I - II, included from males and females. All of the patients had normal cardiac and lung functions with no complications of other systems. Patients were randomized into two groups of 48 in each using a table of random numbers. Patients in the PNS group (treated group) received active electrical stimulation applied at P6, while patients in the control group had an inactive device with similar electrode of the PNS applied at P6 as in the treated group. The patients in the two groups had no significant difference in gender, age, body weight, ASA grade and previous PONV ($P > 0.05$, Table 1).

Methods of Electroacupoint Stimulation

After a qualified and trained acupuncturist determined the site of P6 acupoint on the left forearm and attached a surface electrode (Huawei Ltd., Beijing), one of the rheophores of the PNS (TOF-Watch, Organon Ltd., Dublin, Ireland) was placed at P6. The PNS was turned on to determine the accurate location of the acupoint by eliciting a "de-chi" sensation. Once this was ascertained, the current was adjusted so that the patients could continue to feel the "de-chi" sensation (The stimulator was set at 2-100 Hz, 50 ms, 0.5-4 mA). The electrical stimulation at the acupoints was applied for at least 30 min, but no longer than 60 min before the induction of anesthesia and was continued to the end of surgery.

Anesthesia and Perioperative Management

All patients received fentanyl 50 μ g and midazolam 2 mg by intravenous injection before entering the operating room. The anesthesiologists and care providers were blinded to the study group. Induction of anesthesia was performed by rapid tracheal intubation with propofol 1.5-2 mg/kg and rocuronium 0.6 mg/kg. Isoflurane at 0.5%-1.5%

and supplemental fentanyl, of up to 4 μ g/kg per hour, were used to maintain anesthesia and hemodynamic parameters (heart rate and arterial blood pressure) were kept within 20% of baseline. After the surgery and tracheal extubation, the patients were transferred to the post-anesthetic care unit (PACU). Postoperative data were collected by a separate research nurse who was not aware of the preoperative or perioperative management of patients.

Assessment of Nausea, Vomiting, Antiemetics and Analgesia

The incidence and severity of nausea, vomiting, the use of antiemetics and pain scores were assessed at baseline, 0, 60, 120 min and 24 h post-operation. The subjective experience of nausea and pain at rest was scored using a verbal rating scale (VRS) from 0 to 10, with "0" representing no nausea or pain and "10" representing unbearable nausea or pain. Ondansetron (GlaxoSmithKline, England) 4 mg was intravenously injected as an antiemetic to patients whose nausea score was more than 5, who vomited twice within 15 min or at the patient's request.

Statistical Analysis

SPSS 13.0 software was used to run the statistical analysis. Continuous variables were compared between the two groups using Student's *t* test or two-way ANOVA and descriptive variables were analyzed using χ^2 test. $P < 0.05$ was considered to be statistically significant.

RESULTS

All 96 patients completed the study. As shown in Table 1 and Table 2, the patients' gender, age, weight, ASA physical status, previous PONV history, duration of surgery or anesthesia, transfusion amount, operative procedure and doses of opioids in the two groups were not significantly different.

Results for PONV are given in Table 3 and

Table 2. Comparison of Intraoperative Data between Two Groups ($\bar{x} \pm s$)

Group	Case	Duration of surgery (min)	Duration of anesthesia (min)	Fluids administered (mL)	Intraoperative fentanyl (μ g)
Treated	48	108 $\pm$ 38	136 $\pm$ 37	850 $\pm$ 350	286 $\pm$ 186
Control	48	115 $\pm$ 36	142 $\pm$ 34	900 $\pm$ 420	306 $\pm$ 198

Table 3. Comparison of Postoperative Nausea Scores and Patients with Severe Nausea and Pain between Two Groups

Group	Case	Nausea score ($\bar{x} \pm s$)			Patients with severe nausea [Case (%)]	Patients with severe pain [Case (%)]
		0	60 min	120 min		
Treated	48	1.1 $\pm$ 2.8	0.5 $\pm$ 1.6**	0.6 $\pm$ 2.1**	2 (4.1%)*	9 (18.7%)*
Control	48	1.3 $\pm$ 2.5	2.6 $\pm$ 3.0	1.7 $\pm$ 2.6	14 (29.1%)	26 (54.1%)

Notes: * $P < 0.05$, ** $P < 0.01$, compared with the control group

Table 4. In the first 24 h after surgery, PONV occurred in 14 patients (30%) in the treated group and 31 patients (65%) in the control group. The incidence of PONV was significantly less in the treated group than in the control group ($P < 0.05$). The need for antiemetics was significantly less in the treated group ($P < 0.05$). The degree of nausea and the incidence of severe nausea were significantly lower in the treated group ($P < 0.01$). Similarly, patients in the treated group experienced lower pain scores at 24 h postoperatively and fewer patients in this group had severe pain and severe nausea defined as VRS > 5 of 10 ($P < 0.05$). There was no significant difference in the amount of postoperative analgesics. There were no differences in the duration of PACU stay and no redness, swelling, itching, and pain, or other relevant complications at P6 in the two groups.

Table 4. Incidence of PONV and Requirement for Rescue Antiemetics at 0-24 h [Case (%)]

Group	Case	PONV		Rescue antiemetics of 1 dose
		Nausea	Vomiting	
Treated	48	13 (27.1)*	3 (6.3)	9 (18.7)*
Control	48	32 (66.7)	5 (10.4)	18 (37.5)

Note: * $P < 0.05$, compared with the control group

DISCUSSION

Laparoscopic cholecystectomy for the treatment of cholecystopathy is extensively applied and accepted by both surgeons and patients due to its numerous advantages. The incidence of PONV has been reported to be as high as 53%-80%. After laparoscopic cholecystectomy, the PONV rate could reach 70% if patients do not have any antiemetic prophylaxis⁽²⁾.

It was reported recently that acupuncture may

obviously alleviate pain, PONV and reduce the dose of analgesics during surgery⁽⁷⁾. In the reported study, electroacupoint stimulation with the TOF mode at P6 was significantly effective and better than the control group for the prevention of PONV. Interestingly, patients in the treated group also experienced less pain. Gan, et al. also found that patients in the group of acustimulation at P6 had a better pain relief effect than the placebo group. It has been demonstrated that analgesia produced by acustimulation is related to the body endorphin system and the analgesia can be antagonized by naloxone⁽⁸⁾.

The etiology of PONV after laparoscopic cholecystectomy is not completely understood. Risk factors such as prolonged CO₂ pneumoperitoneum, surgical stimulation of gall bladder, perioperative use of isoflurane, fentanyl and glycopyrrolate and postoperative analgesia with morphine may all contribute to these episodes. CO₂ pneumoperitoneum significantly increases peritoneal pressure, reduces intestinal blood flow and leads to intestinal ischemia and release of emetogenic substances. The gastrointestinal tract manipulation during surgery may stimulate the emetic center. Perioperative hypotension may cause brainstem hypoxia and thus trigger the vomiting center to induce emesis. An important intestinal response to ischemia is the release of serotonin, a strong emetogenic substance^(2,4).

The most probable mechanism of action of acustimulation (TEAS) is attributed to the release of endogenous opioids and the modulation of other *in vivo* neurotransmitters⁽¹⁾. Peripheral nerve stimulation is an integral part of the mechanism as it is not clinically effective if the nerve supply

is disrupted. It was reported that acupuncture stimulation at P6 could increase β -endorphin levels in the human cerebrospinal fluid. It is postulated that opioids may also have an antiemetic effect mediated by the action of β -endorphin on receptors⁽⁹⁾. In addition, the activation of serotonergic and norepinephrinergic fibers may play a PONV-protective role possibly through changes in the serotonin level⁽²⁾. Acustimulation may have a more potent effect on specific neurotransmitters in the pathways linking the vestibular system to the emetic center⁽¹⁾.

There are various drug therapies for the prevention of PONV, including droperidol, transdermal scopolamine, phenothiazines, antihistamines, metoclopramide and ondansetron, etc. They are effective antiemetics, but they are also associated with side effects such as agitation, excessive sedation, extra-pyramidal reactions, restlessness, dystonic reactions, tachycardia and Parkinson-like symptoms⁽²⁾.

Non-pharmacological methods like acupuncture, acupressure, TENS and laser stimulation have shown comparable antiemetic efficacy^(2,5). It was also reported that needling at P6 during surgery may reduce pain, PONV and the need for opioids. One of the most promising non-pharmacologic techniques to prevent PONV is the use of transcutaneous electrical nerve stimulation at P6. The P6 meridian point in acupuncture has been used to treat vomiting and other stomach ailments in traditional Chinese medical practice. Studies show that acupuncture or acupressure at P6 was as effective as a standard antiemetic in the treatment of nausea and vomiting^(2,8,10,11). White, et al. found that the efficacy of needling at P6 was favorably compared to 4 mg ondansetron. Interestingly, the nausea score was lower at 24 h post-surgery in the acustimulation group compared with the ondansetron group⁽⁸⁾. Studies demonstrated that needling at P6 must be conducted before the emetic stimulus to be effective^(5,11) and at a certain stimulation strength⁽¹¹⁾.

In conclusion, transcutaneous electroacupoint stimulation at P6 with the TOF mode not only reduces the incidence and severity of PONV in 24 h following laparoscopic cholecystectomy

but also reduces the antiemetic requirement needed to control PONV. Needling at P6 has also demonstrated postoperative analgesic effects.

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